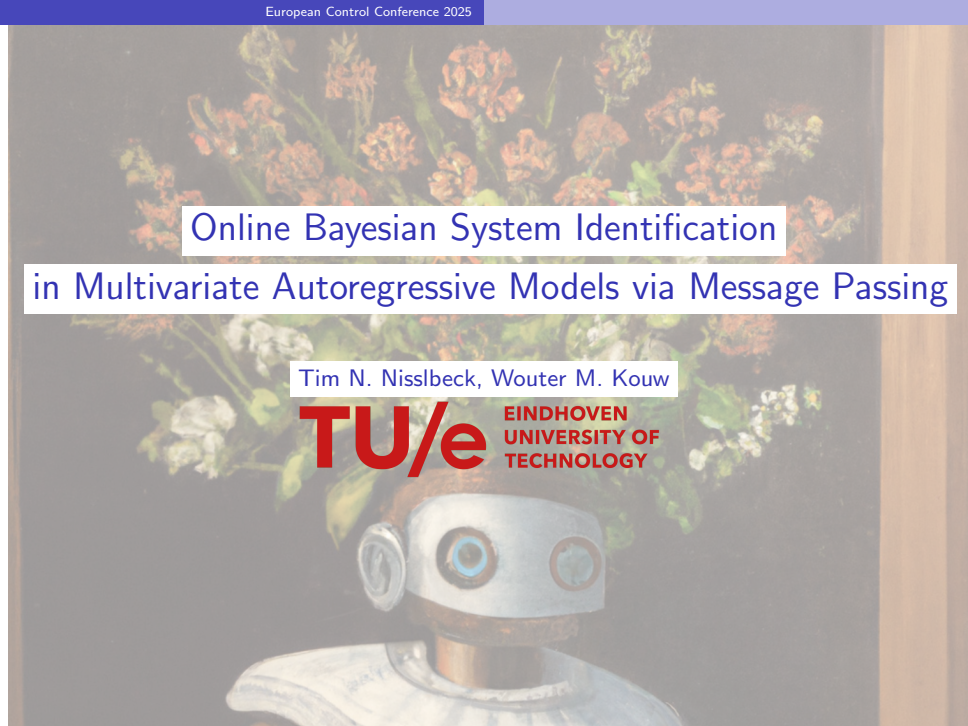


The background of the slide is a painting of a bouquet of flowers, including red and white blossoms, with green leaves. In the foreground, at the bottom center, is a small, stylized robot with a metallic head, large blue eyes, and a silver body.

Online Bayesian System Identification in Multivariate Autoregressive Models via Message Passing

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**EINDHOVEN
UNIVERSITY OF
TECHNOLOGY**

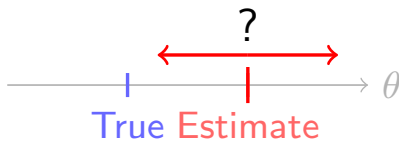
Motivation

Classical ID $\rightarrow$ point estimates



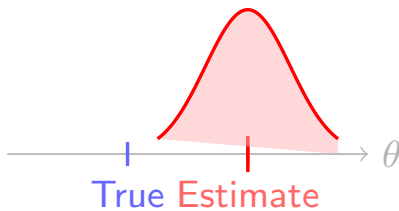
Motivation

Lacks uncertainty quantification



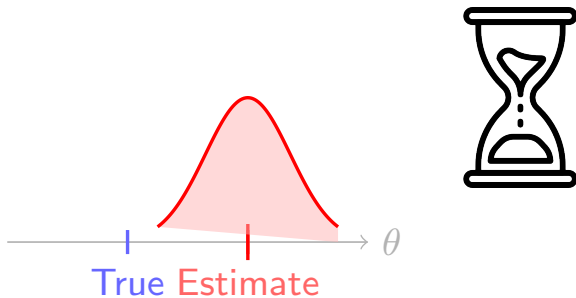
Motivation

Bayesian ID $\rightarrow$ full posteriors



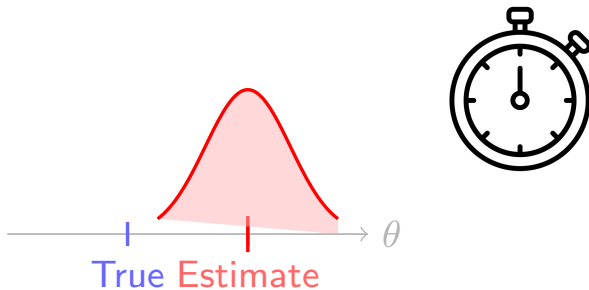
Motivation

Most Bayes methods are offline



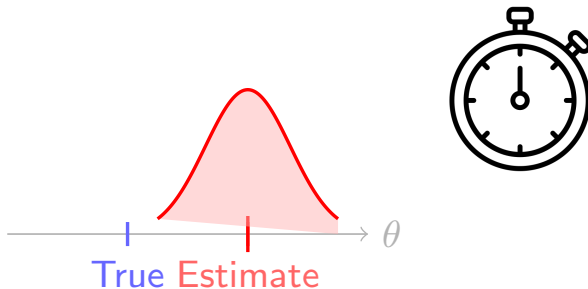
Motivation

Goal: real-time Bayesian ID



Motivation

Fast inference via message passing on factor graphs



Problem

Dynamical systems

We study discrete-time state-space systems defined by:

- **State transition:** $z_t = f(z_{t-1}, u_t)$
- **Measurement:** $y_t = g(z_t) + e_t$

where at time t

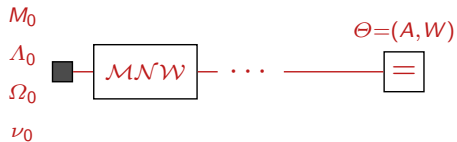
- $z_t \in \mathbb{R}^{D_z}$ state
- $u_t \in \mathbb{R}^{D_u}$ control signal
- $y_t \in \mathbb{R}^{D_y}$ measurement
- $e_t \in \mathbb{R}^{D_y}$ stochastic disturbance

Goal

Predict future outputs y_τ for $\tau > t$ given future inputs u_τ using a model of the system dynamics.

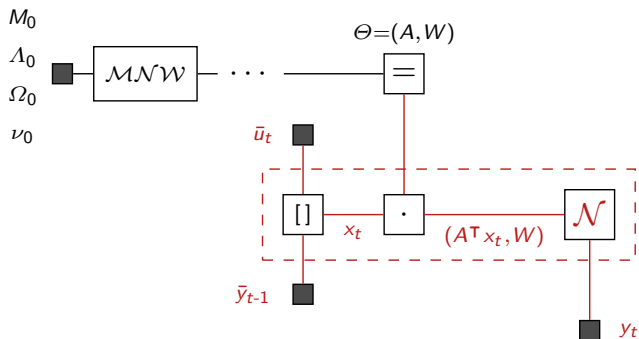
Inference as Message Passing on Factor Graphs

Inference as Message Passing on Factor Graphs



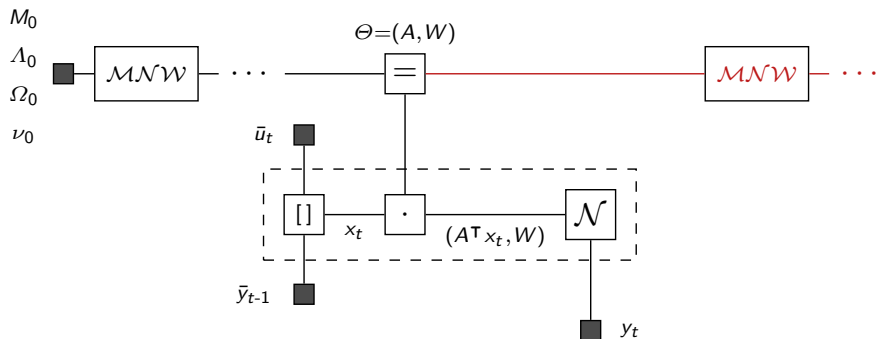
Prior

Inference as Message Passing on Factor Graphs



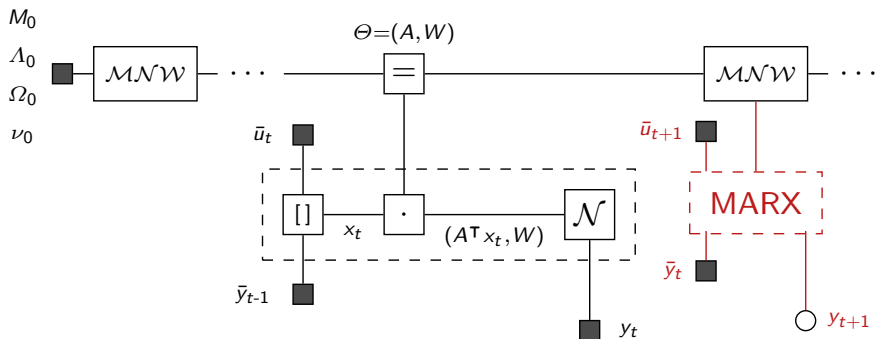
Likelihood

Inference as Message Passing on Factor Graphs



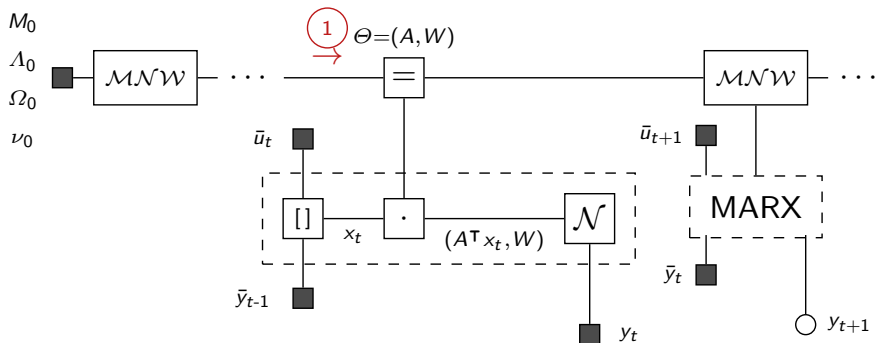
Posterior

Inference as Message Passing on Factor Graphs



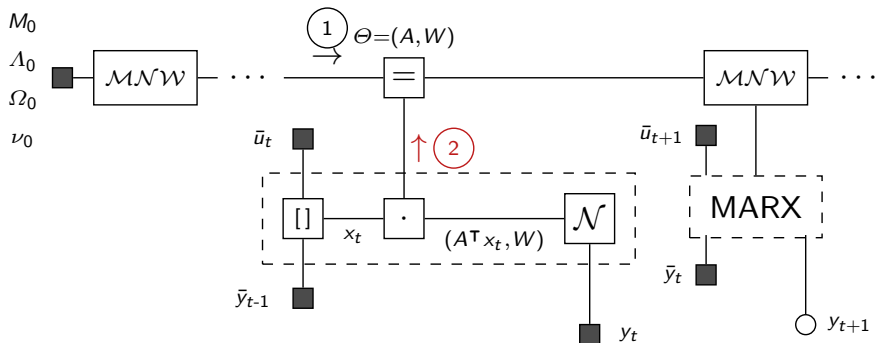
Posterior Predictive

Inference as Message Passing on Factor Graphs



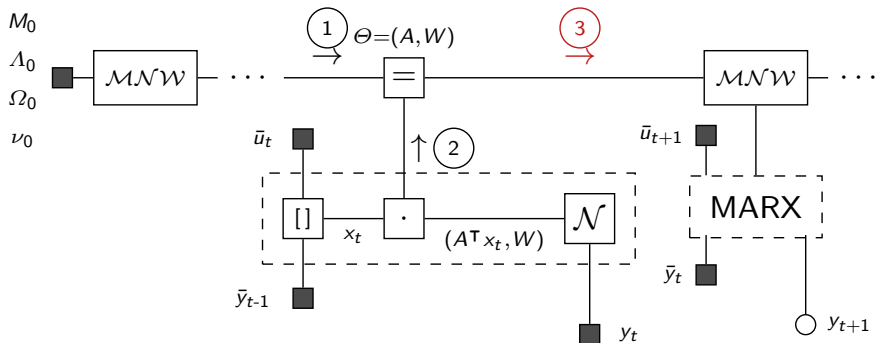
Prior message

Inference as Message Passing on Factor Graphs



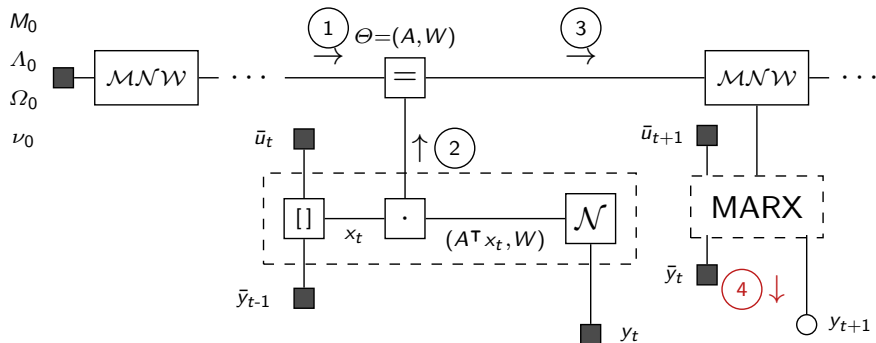
Likelihood message

Inference as Message Passing on Factor Graphs



Posterior message

Inference as Message Passing on Factor Graphs



Posterior predictive message

Experiments

- Verification: Synthetic MARX system with known system parameters

$$y_k = A^T x_t + e_t \quad e_t \sim \mathcal{N}(0, W^{-1})$$

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Experiments

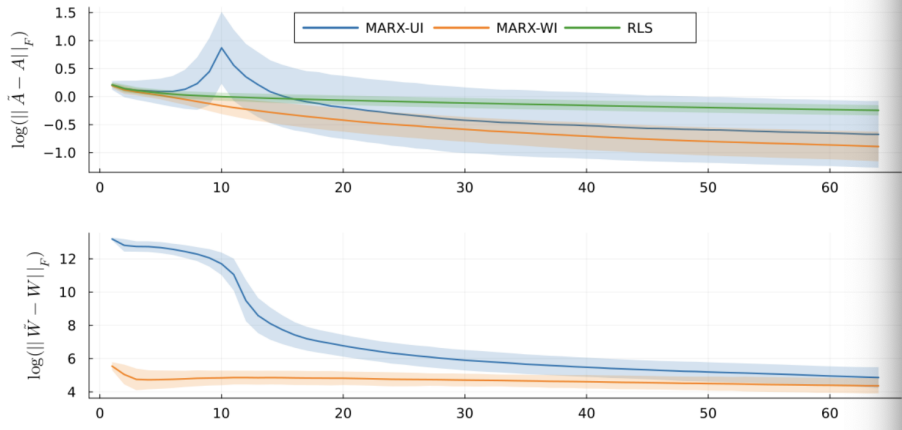
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$$y_k = A^T x_t + e_t \quad e_t \sim \mathcal{N}(0, W^{-1})$$

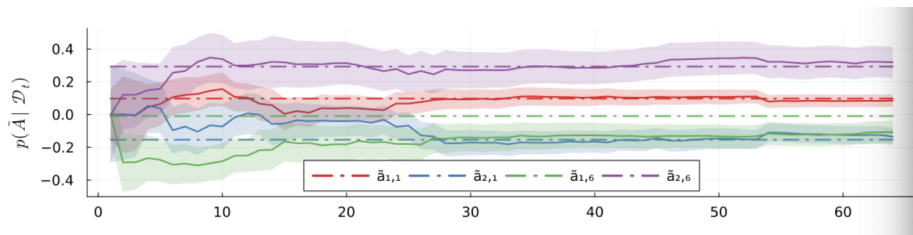
- Baseline: Recursive Least-Squares (RLS)
- Use of two priors: uninformative (MARX-UI) and weakly informative (MARX-WI)

	M_0	Λ_0	Ω_0	ν_0
Uninformative	$0_{D_x \times D_y}$	$1 \times 10^{-4} \cdot I_{D_x}$	$1 \times 10^{-5} \cdot I_{D_y}$	$D_y + 3$
Weakly informative	$0_{D_x \times D_y}$	$1 \times 10^{-1} \cdot I_{D_x}$	$1 \times 10^{-2} \cdot I_{D_y}$	$D_y + 3$

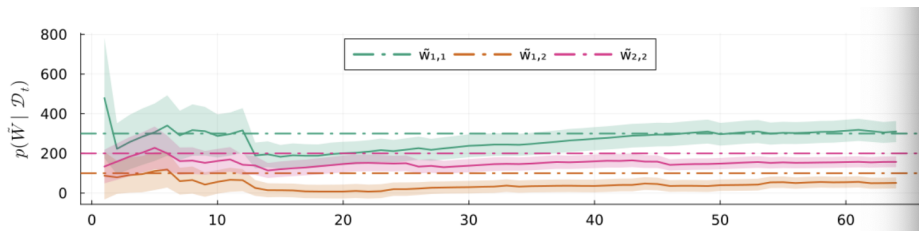
Results



Results



Results



Discussion

- Distributed inference: multi-agent, federated identification

Discussion

- Distributed inference: multi-agent, federated identification
- Modularity of factor graph
 - interpretability
 - extensibility (e.g. hierarchical or time-varying models)

<https://github.com/biaslab/ECC2025-MARXEFE>



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Thank you. Questions?